Fiscal Year:	FY 2018	**Task Last Updated:**	FY 01/11/2018
PI Name:	Wolverton, Scot Ph.D.		
Project Title:	Characterizing Plant Gravity Perception Systems		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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City:	Delaware	**State:**	OH
Zip Code:	43015-2333	**Congressional District:**	12
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	03/16/2015	**End Date:**	03/15/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 3/15/2020 per NSSC information (Ed., 12/17/18) NOTE: Extended to 3/15/2019 per NSSC information (Ed., 12/18/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX15AG55G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plant growth is characterized by plasticity, allowing environmental cues like light and gravity to play a significant role in determining the overall form of the plant body. Statolith sedimentation in specialized cells represents one way that plants perceive gravity. Plants lacking functional statoliths show reduced but consistent gravity responses, suggesting the presence of a non-statolith sensory system. This proposal seeks to investigate the gravity perception machinery in roots of several genotypes of the higher plant Arabidopsis thaliana by taking advantage of the unique microgravity environment of Earth orbit on the International Space Station (ISS). One thrust of the work described here is to characterize the physiological properties of the putative non-statolith sensory system, including the threshold acceleration force required to activate it, and its capacity to modulate downstream growth regulation machinery. The second thrust is to determine whether the two gravity sensory systems control a single or multiple output systems. Addressing this question requires the microgravity environment in order to quantify the kinetics of gravitropism in roots having or lacking functional statoliths under fractional g conditions. The overall outcome of the work proposed here will be an improved understanding of how higher plants perceive gravity and transduce this physical cue into directional growth, a fundamental requirement for improving plants for production on Earth and in long-term spaceflight missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overall outcome of the work proposed here will be an improved understanding of how higher plants perceive gravity and transduce this physical cue into directional growth, a fundamental requirement for improving plants for production on Earth and in long-term spaceflight missions.
Task Progress:	This year we completed all the required ground control experiments to prepare for the flight experiment. We carried out a successful Experiment Verification Test (EVT) in flight hardware and under flight-like conditions at NASA Ames Research Center (ARC). We also built a flight-like set of hardware and carried out an Operations Verification Test (OVT). The OVT took place at the Norwegian User Support and Operation Centre (N-USOC) in Trondheim, Norway from August 21-29, 2017. The Plant Gravity Perception (PGP) flight experiment consists of 3 experiment runs of 5 days each; only the third run was performed during the OVT, with 2 ECs (experiment containers) on each rotor. All steps were performed according to the OVT procedure [AD13]. Sample processing was performed for training purposes at the end of the experiment run. Representatives from NASA ARC, the Principal Investigator (PI) team and the ESA (European Space Agency) PIM were present at N-USOC for the test. The main objective of the OVT was to verify functionality of all European Modular Cultivation System (EMCS) operations including experiment set-up and hardware (including EUE) check out, as well as EMCS schedules, PGP schedules and crew procedures, command procedures, voice loops, and communication tools. Functionality would be verified by satisfactory germination and growth. With regards to hardware, operations and schedules, the Plant Gravity Perception OVT was considered a success. The functionality of several EMCS operations-related procedures could be verified, e.g., EMCS schedules, EMCS command procedures. Some of the crew procedures were also verified during the EC insertion, EC removal, and sample processing. Following the successful conclusion of both of the tests described above, on 3 October 2017, a Science Readiness Review was held at ARC. The goal of this review was to obtain concurrence from the NASA ARC Space Biology Division and approval from the NASA HQ Space Life and Physical Sciences Research and Applications (SLPSRA) Division. The results of the Experiment Verification Tests and the OVT were discussed. PGP was permitted to move forward to flight. The focus of ground experiments this year has been on collecting additional gravitropism data under conditions that more closely mimic the flight with regard to lighting. Our preliminary results show a significant effect of light on the rate of gravitropic response in roots. The non-flight objective of this project is to characterize mutants in other potential non-statolith components of the gravity perception machinery in plants. We have continued this work this year, focusing on mutants in various components of mechanoperception pathways in plants. This year we also proposed an augmentation to our original proposal to GeneLab, with the objective of returning the flight plant samples for gene expression analysis. This augmentation proposal, formally submitted in August 2017, was approved for funding. Thus, we began working on the pilot project for this part of the project, aimed at determining the minimum sample requirements to extract usable RNA for RNA-Seq analysis.
Bibliography Type:	Description: (Last Updated: 05/04/2026)
Abstracts for Journals and Proceedings	Borish A, Fedoush S, Meyer M, Madonich NJ, Wolverton C. "Ground Control Data for a Study of Plant Gravity Perception on the ISS: Weak Ambient Light Enhances Root Gravitropism." Presented at 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Significant Media Coverage	Smola J. (Jennifer Smola, reporter) "Ohio Wesleyan biologist blasting plants into space. Article about PI's research." Columbus Dispatch, Dec 3, 2017. http://www.dispatch.com/news/20171203/ohio-wesleyan-biologist-blasting-plants-into-space , Dec-2017
Significant Media Coverage	Baracy A. (Ashlee Baracy, reporter) "Local professor teams up with NASA to send plants to space. Television news coverage about PI's research." Television news coverage, 10TV Columbus, Dec 4, 2017. https://www.10tv.com/article/local-professor-teams-nasa-send-plants-space , Dec-2017
Significant Media Coverage	Bosco T. (Tom Bosco, reporter) "Which way is up? OWU professor hopes his plants will find the answer in space. ABC6 Columbus local news about PI's research." Local news, ABC6 Columbus, Dec 4, 2017. http://abc6onyourside.com/news/local/which-way-is-up-owu-professor-hopes-his-plants-will-find-the-answer-in-space , Dec-2017

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Significant Media Coverage	NASA Communications. "SpaceX/Dragon CRS-13 What's on Board Science Briefing. YouTube video about the mission including about PI's experiment." NASA TV broadcast, December 11, 2017. https://www.youtube.com/watch?v=p11vBQfPE9E , Dec-2017